Complete F_2/F_3 models and actual degree-two disproofs

Lean checks all 32 binary and 243 ternary general Weierstrass coefficient tuples. Under nonzero discriminant and the exact retained source trace predicate, actual point counts belong to {1,3,5} over F_2 and {1,4,7} over F_3. Every prime divisor has modular embedding degree at most six. Two actual nonsingular curves have subgroup primes 5/7 and exact degrees 4/6, directly disproving universal exact degree two without the characteristic restriction. [pcc:unverified]

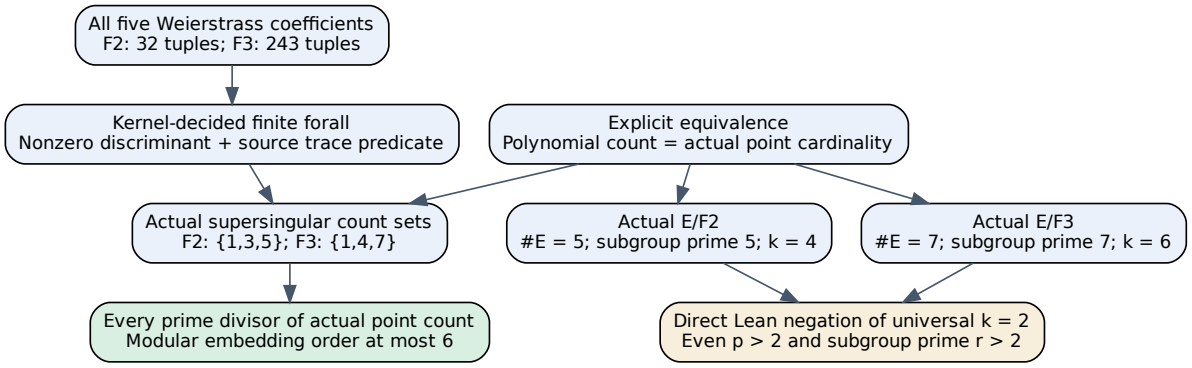


Figure 1: Kernel finite checks, actual point-count bridge and explicit counterexamples. [pcc:unverified]

Actual curve	Field	#E	Trace	Prime r	k
$y^2+y=x^3+x$	F_2	5	−2	5	4
$y^2=x^3-x+1$	F_3	7	−3	7	6

Actual points of orders five and seven exist in the actual Mathlib groups; they are not assumed abstract subgroups. The direct negation still holds with $p>2$ and $r>2$, witnessed by the characteristic-three row. The qualified $p\geq 5$ result is not refuted. [pcc:unverified]

Proof map, assurance and scope

The executable count enumerates polynomial-equation solutions plus the point at infinity. `actual_cardinality_eq_counted` proves an explicit subtype/Option equivalence to the actual Mathlib nonsingular-point set when the discriminant is nonzero. Kernel computation proves the finite universal statements over all five coefficients a_1,a_2,a_3,a_4,a_6. No short-model restriction, random sample or native arithmetic is substituted. [pcc:unverified]

Module	Proved obligation
`PrimeFieldControls`	Actual curves, nonzero discriminants, complete point counts, trace predicate, actual prime-order points, exact modular orders and direct universal negations
`AllPrimeFieldModels`	Generic count equivalence, tuple counts 32/243, closed finite-forall checks, actual count classifications, prime-divisor bound six and combined binding

Main theorem: `SmallCharacteristicCensusAudit.all_prime_field_models_audit_certificate`. Whole gate: 28 public declarations, zero failing and zero trusted-native. Lean 4.35.0 / Mathlib 56a00585 are pinned. Only propext, Classical.choice, Quot.sound; no holes, custom axioms, native_decide exception, linter waiver or added elaboration setting. Gate receipts and all 11 intermediate source/log files are archived with byte-equal unpack checks. [pcc:unverified]

The source predicate is exactly `Int.gcd (p+1−Nat.card E.Point) p ≠ 1`; modular embedding degree is `orderOf (p : ZMod r)`. This does not silently identify modular order with rationality of the full r-torsion. The all-model result covers extension degree one over F_2 and F_3 only. Geometric supersingularity equivalence, higher extension fields and the full torsion-field bridge remain separate. The general Hasse/classification proof is not inferred from these fields. [pcc:unverified]

Source commit `c1ca2f5f` precedes theorem packet `PCC-20261110-2661fc`. Full local verification passes at 10:10:52Z; signed standing remains unverified. Sources and actual verdict/standing are recorded in `../../sources.json` and `../../verification.json`. The first Cauchy wrapper failed because its classical equality choice differed from the caller's computable point-group instance; an explicit equality parameter repairs the proof without changing the group or claim. Earlier linter warnings were removed without waivers; unsuccessful sources/logs remain archived. [pcc:unverified]

Sources checked: Freeman–Scott–Teske (2010), Definition 2.1/Remark 2.2 pp 230–231 and trace-square list p 237; Waterhouse (1969), Theorem 4.1 pp 536–537, with prior visual/read receipts retained in census source plans. Mathlib affine equation/nonsingular-point equivalence, finite-order and Cauchy declarations were read. KB/Conductor unavailable; bounded primary, repository and Mathlib fallback. No novelty, timing, native census, sampling, security-cost or paid-compute claim. The original fourteen-item audit remains partial. [pcc:unverified]